## VISUAL BASIC DESKTOP APPLICATION PROJECT RUBRICS

VB.NET Windows Forms | Microsoft Access Database

Total: 100 Points

This rubric is designed for evaluating students' final desktop application projects developed using Visual Basic .NET (Windows Forms) with Microsoft Access as the database, emphasizing complete system functionality, database integration, reports generation, data validation, and account security.

### Grading Criteria

| Criteria | Description | Points | SCORE |
|---|---|---|---|
| 1. System Functionality | All proposed modules and features are operational, including Add, Edit, Delete, View, Search, and other required processes. No major errors or broken functions. | 25 | |
| 2. VB.NET & Microsoft Access Database Integration | Proper use of VB.NET programming, Microsoft Access database connectivity (OleDb), CRUD operations, and accurate data storage and retrieval. | 15 | |
| 3. Reports Generation | Generates accurate and complete reports using actual database records, with appropriate filters, summary calculations, and printing or export functionality (PDF/Excel). | 15 | |

| | | | |
|---|---|---|---|
| 4. Input Validation & Error Handling | Implements required-field validation, correct data formats, duplicate prevention, appropriate error messages, and proper exception handling using Try-Catch. | 15 | |
| 5. Account Credentials & System Security | Secure login/logout, password hashing, role-based access, protection against unauthorized access, parameterized queries, and secure handling of database files. | 15 | |
| 6. User Interface & User Experience | Professional Windows Forms design, consistent layout, clear navigation, properly labeled controls, and user-friendly forms. | 10 | |
| 7. Project Presentation & Demonstration | Clear explanation of the system, successful live demonstration, and ability to answer technical questions about VB.NET code, Microsoft Access database, and security. | 5 | |
| **Rating** | **Performance Level** | **Percentage of Criterion** | **SCORE** |
| 5 | Excellent – Fully functional, complete, and error-free | 100% | |

| | | | |
|---|---|---|---|
| 4 | Very Good – Functional with minor issues | 85% | |
| 3 | Satisfactory – Functional but with noticeable limitations | 70% | |
| 2 | Needs Improvement – Several incomplete or malfunctioning features | 50% | |
| 1 | Poor – Mostly incomplete or nonfunctional | 25% | |
| 0 | Not Implemented – Feature is missing | 0% | |
| TOTAL | | 100 | |

Scoring Formula:

\text{Points Earned}=\text{Criterion Points}\times\text{Rating Percentage}

## Mandatory Technical Requirements

Student Project Evaluation Checklist

0 of 12 verified

Reset

VB.NET Windows Forms application runs successfully

Microsoft Access (.accdb) database connection is functional

Complete Create, Read, Update, Delete (CRUD) operations

Working login and logout functionality

Passwords are securely hashed and verified, not stored in plain text

User roles and access permissions are enforced

Input validation for required fields, data types, and duplicates

Parameterized OleDb queries to prevent SQL injection

Try-Catch error handling with appropriate user messages

Reports generated from actual Microsoft Access records

Reports support filtering and printing or exporting

Database file access is restricted to authorized users

## Important Grading Policies

- Security: Plain-text passwords, hardcoded administrator credentials, and unrestricted access to protected forms are critical deficiencies.

- Database: Students must demonstrate actual Microsoft Access integration. Hardcoded records or sample data displayed without database connectivity are insufficient.
- Validation: Required-field checks, duplicate prevention, and data-type validation must be implemented in the VB.NET application.
- Reports: Generated reports must reflect actual database records, not manually prepared or hardcoded results.
- Functionality: All required buttons and modules must perform their intended operations during the live demonstration.
- Database protection: Students should demonstrate appropriate file permissions and explain that desktop login controls alone cannot protect an Access database if users can directly open or copy the file.

## Instructor's Evaluation Sheet

Student / Group Name

Project Title

Select a rating from 0 to 5 for each criterion.

System Functionality

Maximum 25 points

0 – None

—

VB.NET & Access Integration

Maximum 15 points

0 – None

—

Reports Generation

Maximum 15 points

0 – None

—

Validation & Error Handling

Maximum 15 points

0 – None

—

Account Security

Maximum 15 points

0 – None

—

Windows Forms UI/UX

Maximum 10 points

0 – None

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Presentation

Maximum 5 points

0 – None

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TOTAL SCORE

**0.00 / 100**

Copy Evaluation

Suggested passing standard: 75/100, provided the application has functional VB.NET and Microsoft Access integration, operational core modules, and no critical account security vulnerabilities.

This rubric is suitable for a Final Project, Desktop Application Defense, or Practical Examination in Visual Basic .NET with Microsoft Access Database.